100 Days of Linux - Day 028

Title: Understanding Linux File Permissions

Today's Goal
Learn how Linux file permissions work and how to view them using **ls -l**.

Why This Matters
Every Linux system uses permissions to control who can read, modify, or execute files. Understanding permissions is essential for developers, data engineers, and DevOps engineers.

Command of the Day: ls -l

Syntax: ls -l
Explanation: Displays files with detailed information including permissions, owner, group, size, and modification date.

Permission Symbols:
r = Read
w = Write
x = Execute

Permission Groups:
User (Owner)
Group
Others

Example:
-rw-r--r-- 1 user user 120 Jul 14 notes.txt

Beginner Lesson
Every file and directory has permissions. The first character shows whether it's a file (-) or directory (d). The next nine characters are split into three groups: owner, group, and others. Today you'll learn to read these permissions. Tomorrow you'll change them using chmod.

Practice Questions

1. Create a directory named permissions_practice and move into it.

2. Create files script.sh and notes.txt. Run ls -l and observe their permissions.

3. Identify which permissions belong to the owner, group, and others for notes.txt.

4. Create another directory named data and run ls -l again. Compare the permissions of files and directories.

5. Display your current working directory after completing today's tasks.

Hints

Use mkdir, touch, cd, pwd, and ls -l together. Focus on reading permissions, not changing them.

Mini Challenge

Create files app.py, pipeline.py, and config.txt. Use ls -l to

compare their permissions and write down which permissions each has.

Common Mistakes

Confusing the owner with the group or expecting directories to have the same behavior as files.

Did You Know?

Directories need the execute (x) permission to allow users to enter them with the cd command.

Pro Tip

Before changing permissions with chmod, always check the current permissions using ls -l.

Answer Key (Instructor Only)

Q1: mkdir permissions_practice && cd permissions_practice
Q2: touch script.sh notes.txt && ls -l
Q3: Read the three permission groups from ls -l output
Q4: mkdir data && ls -l
Q5: pwd

Homework

Create five files and three directories. Use ls -l to compare their permissions and note the differences.

Next Day Preview

Tomorrow you'll learn chmod to change read, write, and execute permissions.